

QCE Chemistry Units 3&4

Paper 2

Student's Name: _____

Teacher's Name: _____

Time allowed

- Perusal time – 10 minutes
- Working time – 90 minutes

General instructions

- Answer all questions in this question and response booklet.
- Write using black or blue pen.
- QCAA-approved calculator permitted.
- Formula and data booklet provided.
- Planning paper will not be marked.

Section 1 (65 marks)

- 8 short response questions

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2021 QCE Chemistry Units 3&4 Written Examination.

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SECTION 1

Instructions

- If you need more space for a response, use the additional pages at the back of this booklet.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. See page ...
 - If you do not do this, your original response will be marked.
-

DO NOT WRITE ON THIS PAGE

THIS PAGE WILL NOT BE MARKED

QUESTION 1 (7 marks)

A synthesis experiment was conducted in which 4.50 g of $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ ($M = 88.0 \text{ g mol}^{-1}$) was heated with an excess of $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ ($M = 60.0 \text{ g mol}^{-1}$) in the presence of a sulfuric acid catalyst. The mass of the ester ($M = 130 \text{ g mol}^{-1}$) produced was 5.32 g.

- a) Apply IUPAC rules to name the ester produced in this reaction. [1 mark]

- b) Calculate the atom economy for this reaction. Show your working. [2 marks]

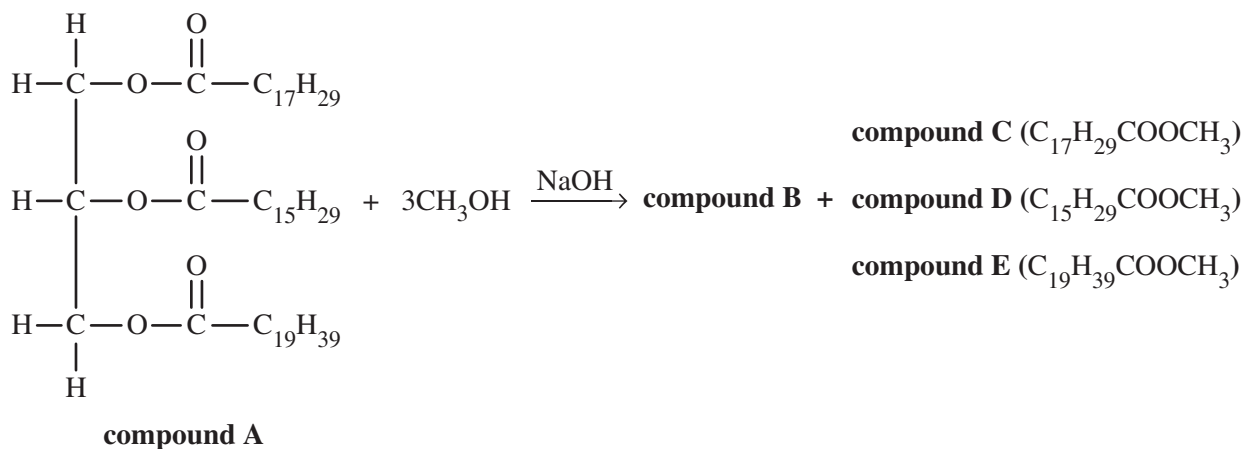
Atom economy = _____ % (to three significant figures)

- c) Calculate the percentage yield for the ester synthesis. Show your working. [4 marks]

Percentage yield = _____ % (to three significant figures)

QUESTION 2 (7 marks)

In the presence of a sodium hydroxide (NaOH) catalyst, a transesterification reaction will occur between methanol and compound A to produce compound B and a mixture of methyl esters (C, D and E). The mixture of methyl esters is known as biodiesel.



- a) i) Identify the type of compound shown as compound A. [1 mark]
- _____
- ii) Deduce the molecular formula of compound B. [1 mark]
- _____
- iii) State which of the methyl esters (compound C, D or E) is derived from a polyunsaturated fatty acid. [1 mark]
- _____
- b) When the transesterification process is conducted, the biodiesel must be extracted from the reaction mixture. To extract the biodiesel, a series of washing steps are used. The biodiesel remains in the upper organic layer, while the NaOH and unreacted methanol move into the lower aqueous layer.
- i) In terms of the structure and bonding, explain why the biodiesel is **not** soluble in the aqueous layer of the washing process. [2 marks]
- _____
- _____
- _____
- _____

- ii) In terms of the structure and bonding, explain why methanol is soluble in the aqueous layer of the washing process.

[2 marks]

QUESTION 3 (8 marks)

Acrylic acid ($\text{CH}_2\text{CHCO}_2\text{H}$) is a precursor for many important plastics. The value of K_a for acrylic acid is 5.6×10^{-5} .

- a) Draw the structural formula for a molecule of acrylic acid. *[1 mark]*

- b) Calculate the pH of a 0.10 M solution of acrylic acid. Include the equation for the ionisation of acrylic acid, and state any assumptions made during your calculation. *[5 marks]*

pH = _____ % (to two significant figures)

- c) 20.0 mL of a 0.10 M solution of acrylic acid was diluted by the addition of 5.0 mL of water.

Will the pH of the solution increase or decrease following the dilution? Explain your reasoning, including reference to any shift in the position of the ionisation equilibrium reaction.

[2 marks]

QUESTION 4 (10 marks)

Molecular components of food may occur in various structural forms that differ in their properties.

- a) i) Monosaccharides may be classified as aldoses or ketoses.

Which of these classifications applies to glucose?

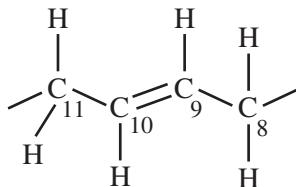
[1 mark]

- ii) Polymerisation of glucose molecules produces two forms of starch – amylose and amylopectin.

Compare amylose and amylopectin in terms of their structure and bonding.

[4 marks]

- b) Fatty acids with the molecular formula $C_{17}H_{33}COOH$ occur as two geometric isomers. Each of these compounds has one double bond at the ninth carbon in the chain. A section of a fatty acid chain with the double bond on the ninth carbon is shown below.



- i) Is this section the *cis* or *trans* isomer?

[1 mark]

- ii) Draw the arrangement of the same atoms and bonds in the other fatty acid isomer.

[1 mark]

- c) Another fatty acid containing 18 carbon atoms has the formula $C_{17}H_{35}COOH$.

Describe a chemical test that could be used to distinguish between the two fatty acids $C_{17}H_{33}COOH$ and $C_{17}H_{35}COOH$, and outline the expected results of the test.

[3 marks]

QUESTION 5 (8 marks)

The boiling points of three compounds are shown in the table below. The three compounds are all colourless liquids at room temperature.

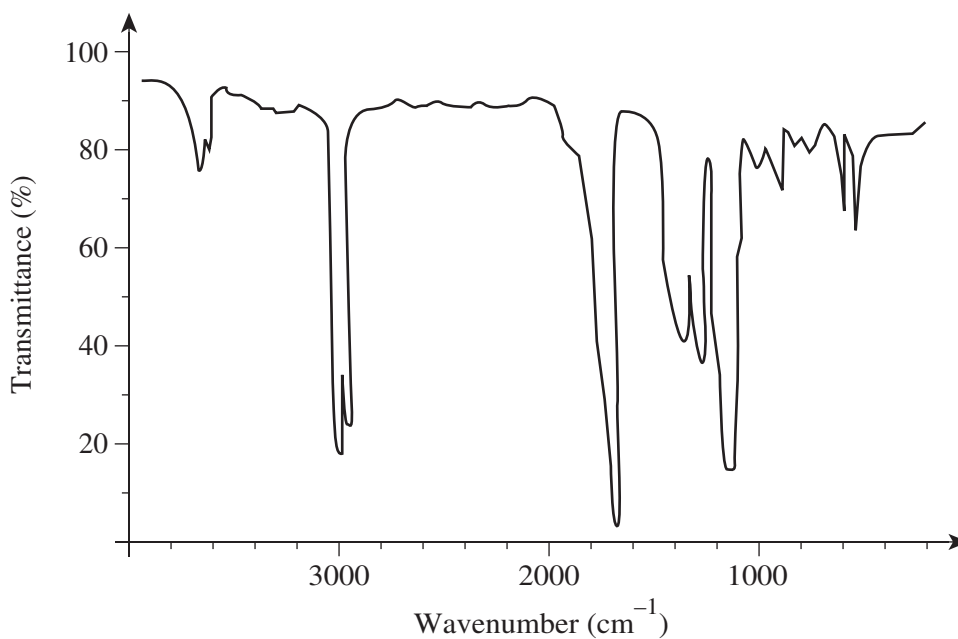
Compound	ethanoic acid	butan-1-ol	butyl ethanoate
Boiling point (°C)	118	118	127

- a) Outline how a simple chemical test could be used in a school laboratory to distinguish between ethanoic acid and butan-1-ol. *[2 marks]*

- b) Draw the structure of the ester, showing all bonds. *[1 mark]*

- c) Even though butan-1-ol is a smaller molecular compound than butyl ethanoate, their boiling points differ by only a small amount.
Explain this observation in terms of the structure and bonding of the compounds. *[3 marks]*

- d) Butyl ethanoate is produced by reacting butan-1-ol and ethanoic acid. The reactants are heated under reflux with concentrated sulfuric acid as the catalyst. Following this reaction, a sample was isolated from the reaction mixture and produced the infrared spectrum shown below.

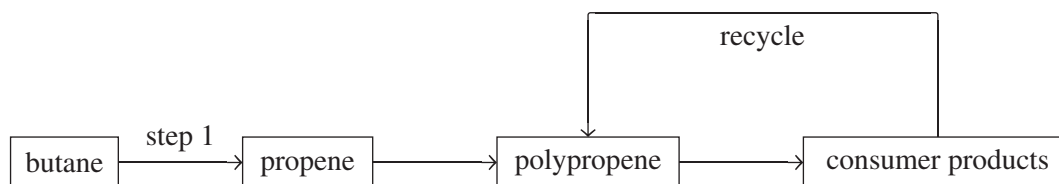


Analyse the data provided to show that the sample is **not** contaminated by either of the reactants.

[2 marks]

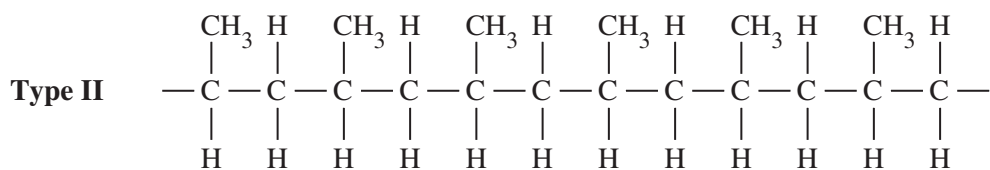
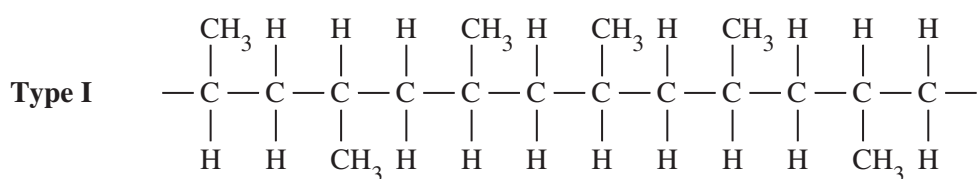
QUESTION 6 (8 marks)

The main steps in the production of polymer products are shown in the flow chart below.



- a) Give the formula of a compound from the same homologous series as butane that contains seven carbon atoms per molecule. *[1 mark]*

- b) Polypropylene is a linear polymer. Two forms of the polymer (type I and type II) are shown below.



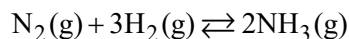
Consider the polymer structures and their intermolecular bonding to deduce which type (I or II) is likely to melt at a higher temperature. Explain your reasoning. *[3 marks]*

- c) i) Discuss one advantage of the use of polymer materials such as polypropene. [2 marks]

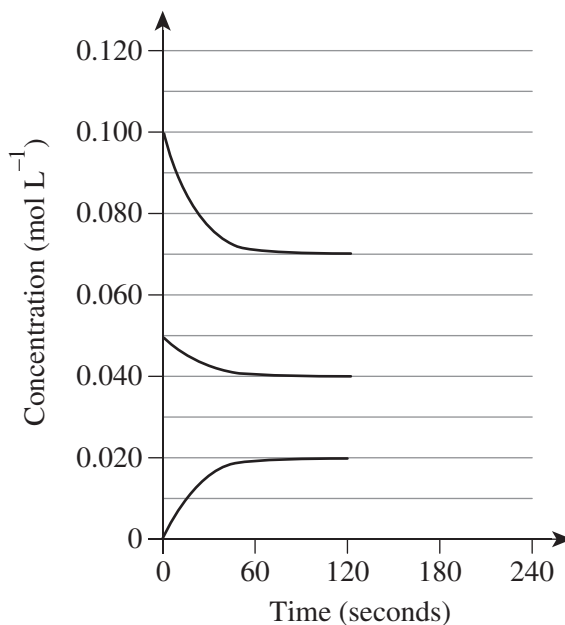
- ii) Discuss one disadvantage of the use of polymer materials such as polypropene. [2 marks]

QUESTION 7 (9 marks)

0.100 mol of $\text{H}_2(\text{g})$ and 0.050 mol of $\text{N}_2(\text{g})$ were placed in a sealed 1.0 L container and allowed to come to equilibrium according to the following equation.



The results from the experiment are shown in the graph below.



- a) i) Write the equilibrium expression, K_c , for the reaction. [1 mark]

- ii) Using the data from the graph, calculate the value of the equilibrium constant for the reaction. [2 marks]

Equilibrium constant = _____ M^{-1} (to two significant figures)

- b) i) At 120 seconds, the volume of the container was changed to 2.0 L, while maintaining a constant temperature.

On the axes above, show the change to the concentration of ammonia at 120 seconds.

[2 marks]

- ii) Complete the table below to show the expected changes as the system returns to equilibrium between 120 and 180 seconds. Indicate your choices by placing a tick in one box in each column.

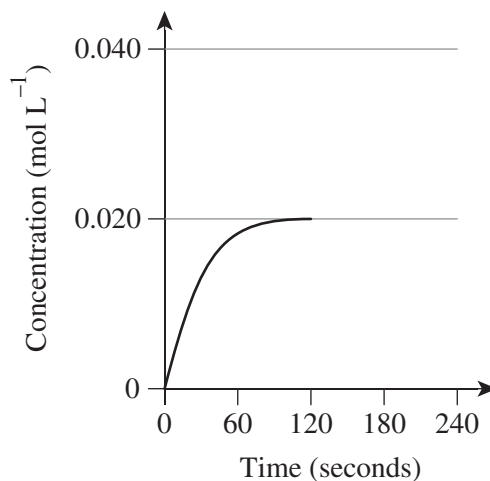
[2 marks]

Shift in equilibrium position		Effect on the value of K_c	
to the product side	☐	increases	☐
to the reactant side	☐	decreases	☐
no change	☐	no change	☐

- c) The original experiment was repeated under identical conditions except that a catalyst was present in the container.

On the enlarged graph below, sketch the effect of the catalyst on the concentration of NH_3 between 0 and 120 seconds.

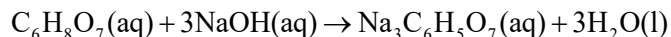
[2 marks]



QUESTION 8 (8 marks)

Lemon juice contains several acidic chemicals, including citric acid and vitamin C (ascorbic acid).

- a) A student determined the citric acid content of a lemon juice sample using an acid–base titration. Citric acid is a weak, triprotic acid and reacts with sodium hydroxide according to the following equation.



2.00 mL of lemon juice was diluted to a total volume of 20.0 mL with distilled water. This 20.0 mL solution was titrated against a standardised 0.102 M sodium hydroxide solution. The average titre required was 14.6 mL. Note that $M(\text{C}_6\text{H}_8\text{O}_7) = 192 \text{ g mol}^{-1}$.

- i) Calculate the concentration of citric acid in the undiluted lemon juice. [5 marks]

Concentration = _____ % m/v (to three significant figures)

- ii) During the titration, the student incorrectly rinsed their burette with deionised water.

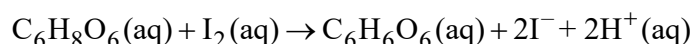
Deduce which of the following would occur as a result of the student's error.

Indicate your response by placing a tick in one of the boxes.

[1 mark]

The calculated level of citric acid would be higher than the true value.	☐
The calculated level of citric acid would be lower than the true value.	☐
There would be no effect on the calculated level of citric acid.	☐

- b) The level of vitamin C in lemon juice can be determined using a redox titration with iodine. The reaction produces dehydroascorbic acid and iodide ions as its products, and starch is used as an indicator. The equation for the reaction is as follows.



Deduce the oxidation half-equation for the reaction.

[1 mark]

- c) Assuming the titration is performed correctly, suggest why the result for citric acid concentration obtained by the acid–base titration described in a) will be an overestimate of the actual level of citric acid in the lemon juice. *[1 mark]*

END OF PAPER

